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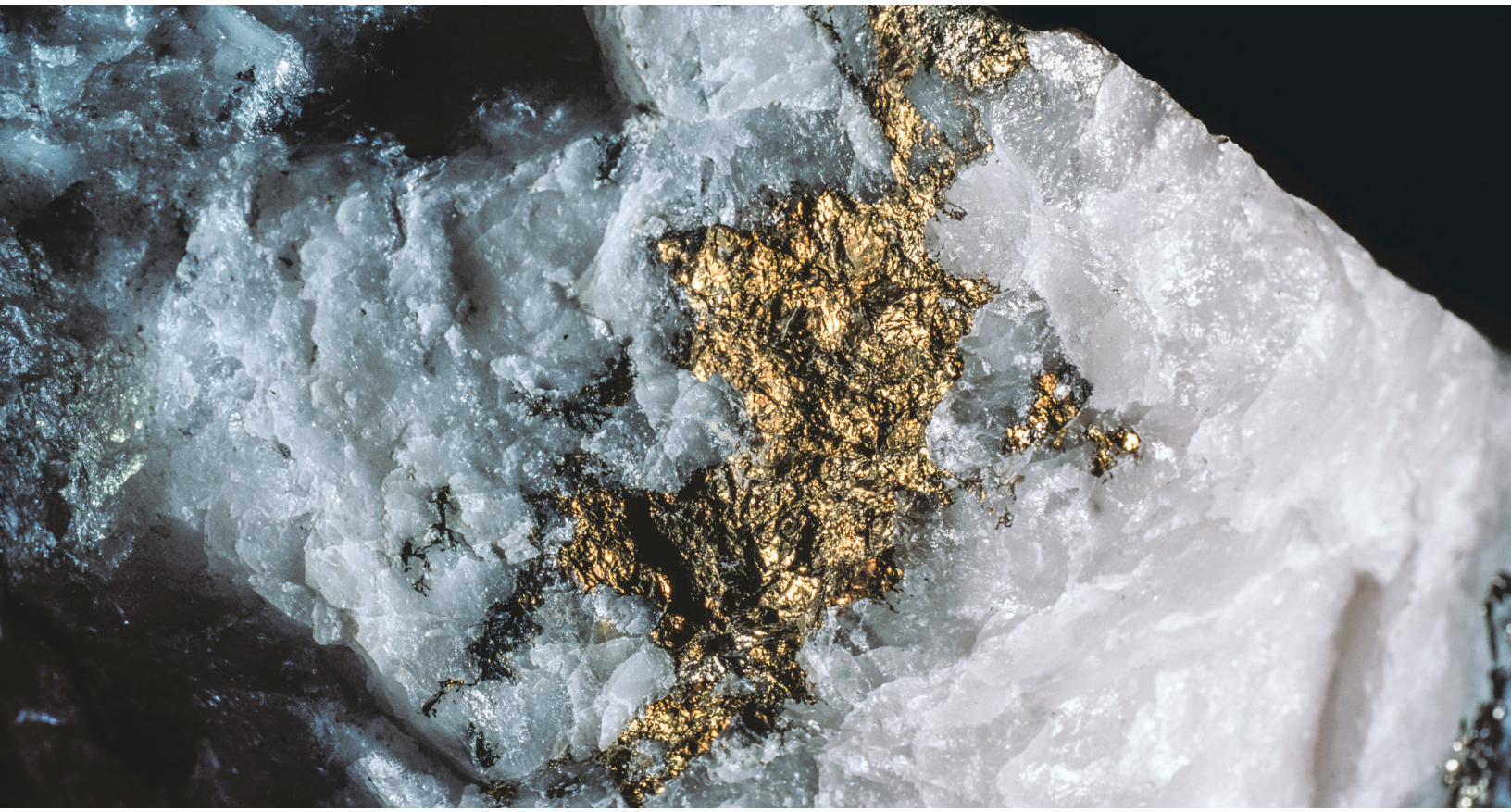
MineSpans Commentary

Refractory gold ores: Challenges and opportunities for a key source of growth

High-grade refractory gold reserves offer the potential for lower costs per ounce despite higher capital and operational costs. A focus on three action areas can help maximize the opportunity.

by Gabriel Motta, Michal Polcyn, and Elijah Saragosa



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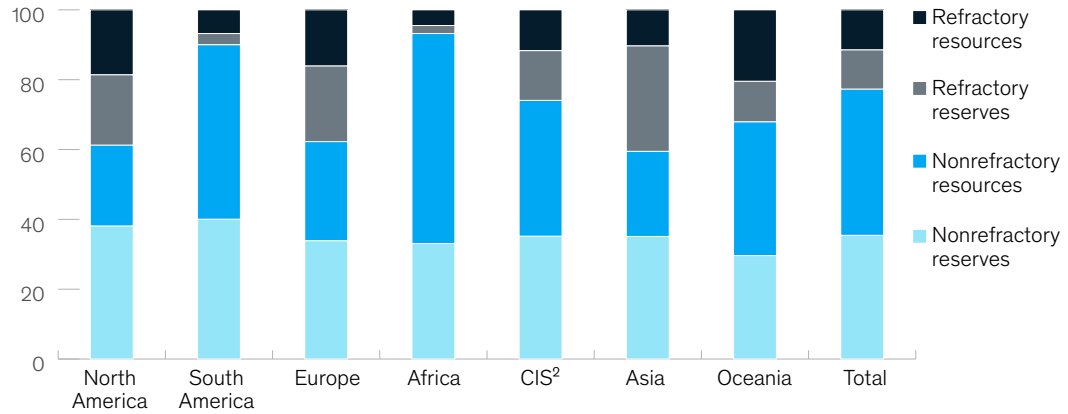
March 2021

Gold miners are facing a reserves crisis, and what is left in the ground is becoming more and more challenging to process.[1] Refractory gold reserves, which require more sophisticated treatment methods in order to achieve oxide-ore recovery rates, correspond to 24 percent of current gold reserves and 22 percent of gold resources worldwide (Exhibit 1). Despite offering a higher grade, these ores can only be processed using specific pretreatment methods such as ultrafine grinding, bio oxidation, roasting, or pressure oxidation (POX). This special treatment is required for two reasons: first, to liberate gold particles encapsulated in sulfide or arsenic minerals; and, second, to eliminate carbonaceous material occurring in the ore, which adsorbs dissolved gold

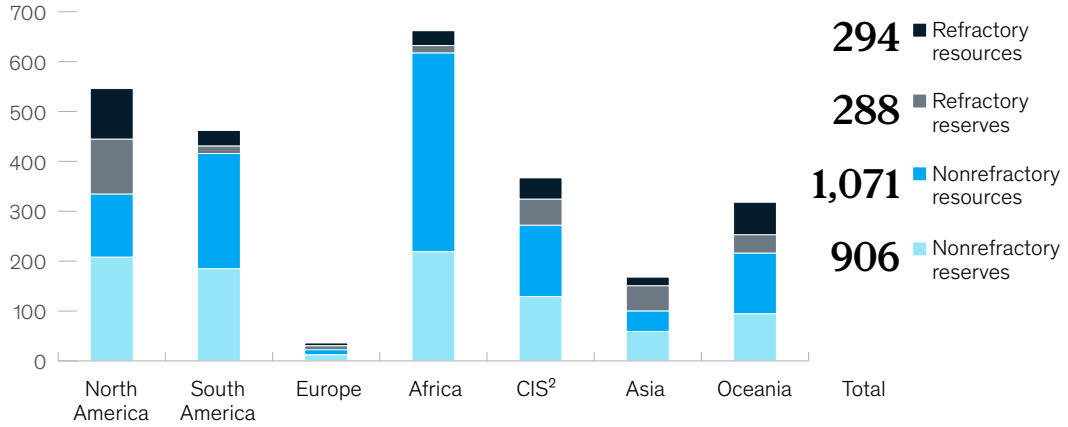
Exhibit 1

Approximately one-quarter of the gold in geological reserves and resources can be considered refractory.

Gold contained in reserves and measured and indicated resources,[1] % share



Gold contained in reserves and measured and indicated resources,[1] million troy ounces

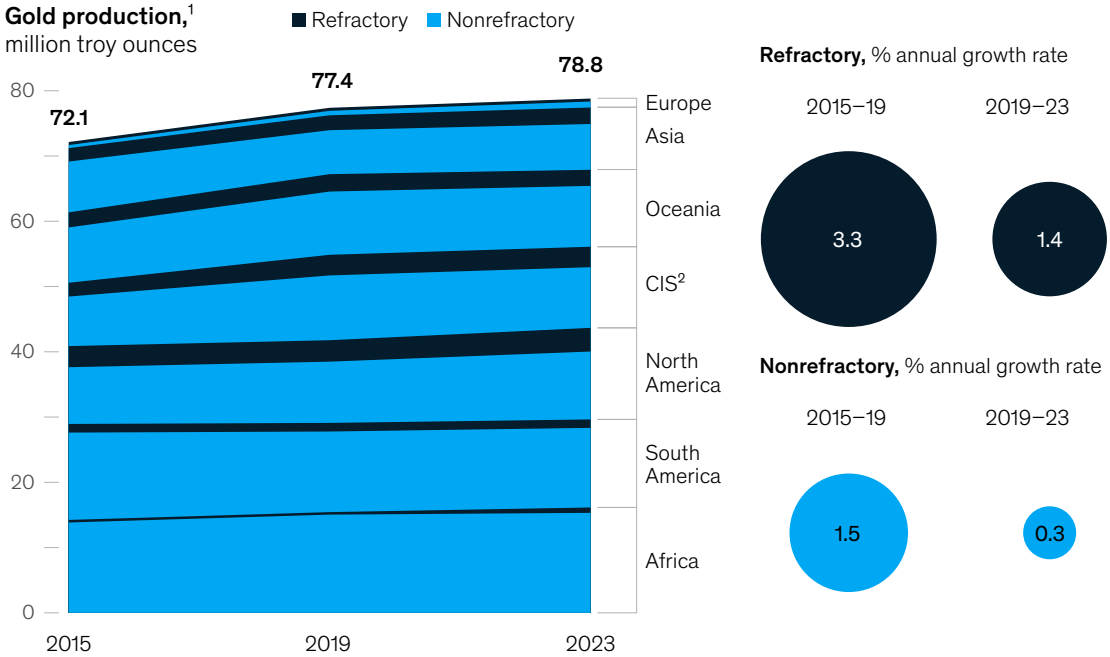


[1]MineSpans covers approximately 75% of global gold production. Excludes artisanal and small-scale producers. Primary gold mines only. Resources exclusive of reserves.
[2]Commonwealth of Independent States.
Source: MineSpans

[1] To learn more, see Greg Callaway and Oliver Ramsbottom, *Can the gold industry return to the golden age?* April 2019, McKinsey.com.

Exhibit 2

Gold production from refractory ores is expected to grow 1.4 percent in the next four years, versus 0.3 percent for nonrefractory ores.



[1]MineSpans covers approximately 75% of global gold production. Excludes artisanal and small-scale producers.
[2]Commonwealth of Independent States.
Source: MineSpans

instead of active carbon that is normally added to the leaching solution.

As most refractory reserves and resources lie in jurisdictions with low mining-regulatory and political risk, risk-averse companies may instead face geological risks associated with refractory-type deposits.

According to MineSpans analysis, approximately one-quarter of the gold in geological reserves and resources can be considered refractory, and most is located in regions with a long history of gold exploration and mining, as well as a lower investment risk, such as North America, Oceania, and the Commonwealth of Independent States (CIS). It is important to note that the additional processing steps required for treating refractory ores generate additional costs compared with conventional plants; however, the reserve grade for these ores is on average 86 percent higher than those of nonrefractory-type deposits (2.25 grams per metric ton on average, versus 1.21 grams per metric ton for nonrefractory ores).

Our analysis shows that, in the near future, production from refractory-type deposits is expected to grow at a higher rate than production from nonrefractory ores (Exhibit 2). This production growth for refractory ores can be explained by analyzing two main factors: costs and grade.

Why refractory ores are growing: Significantly higher grades should continue to offset higher costs per metric ton in the near future

Our analysis of recently developed and planned gold projects for refractory and nonrefractory ores found that:

— *Capital costs per metric ton of ore capacity are higher in refractory-ore projects.* Construction of processing plants with POX circuits (the technology that recently became the most popular to treat difficult ore) requires approximately 48 percent higher investments compared to plants with regular tank-leaching processes (Exhibit 3). Recently constructed POX facilities in Russia and Turkey had a price tag of nearly $1 billion, and the construction of other facilities, which are expected to cost more than $2.5 billion, are still in the pipeline.

— *Operational costs per metric ton of processed ore are higher on average.* Operational costs vary depending on the mining method and is notably 50 percent higher for open-pit refractory-ore projects. MineSpans data analysis shows that the increase in operational costs is primarily driven by higher consumables and energy costs (Exhibit 4).

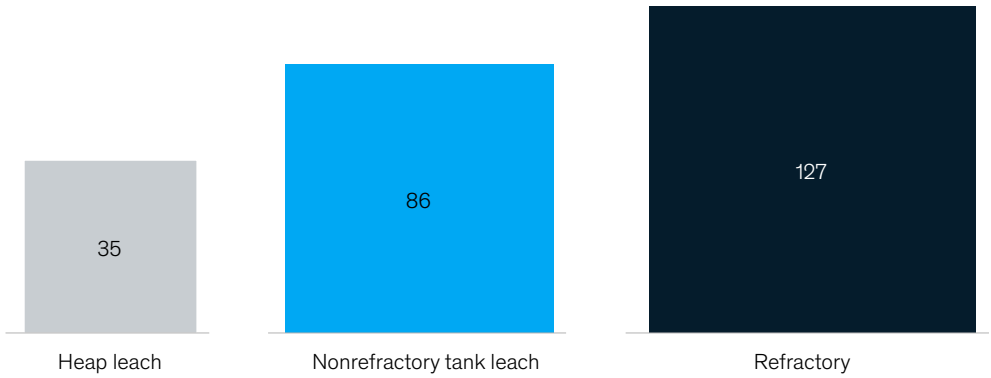
— *Due to their significantly higher grades, refractory ores yield costs per ounce that are frequently lower than the average costs for conventional ores.* Mill-head grades of refractory deposits can be 86 percent higher; as a consequence, the operational costs per ounce of gold produced are approximately 19 percent lower in the case of refractory gold mines.

Thus, according to MineSpans data, 54 percent of gold production from refractory deposits comes from mines situated in the bottom half of the cost curve, while only 18 percent sits in the fourth quartile (Exhibit 5). This high-grade effect is expected to remain in place at least until 2023, but grade erosion should dampen it over time.

Exhibit 3

Up-front project capital-expenditure intensity of open-pit mines varies depending on ore-processing method.

Project capital-expenditure intensity of open-pit mines, $ per metric ton of run-of-mine (ROM) capacity

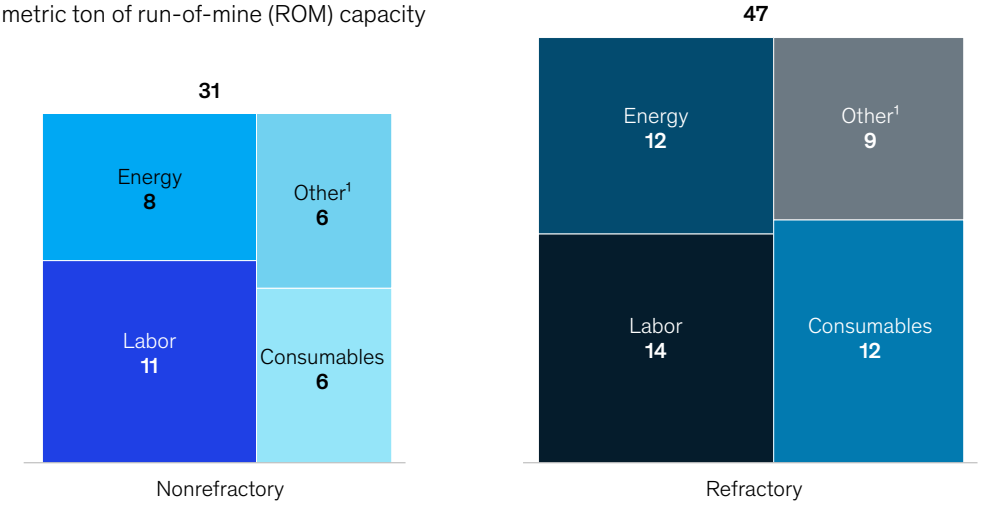


Source: MineSpans

Exhibit 4

Operational cost increases for open-pit mines are driven by higher consumables and energy costs.

Operational costs of open-pit mines with split by cost factors,
$ per metric ton of run-of-mine (ROM) capacity



[1]For example freight, mine overheads, etc.
Source: MineSpans

How to maximize the opportunity: Mine-design optimization, digital-technology application, and asset-portfolio planning

In order to generate the most value from refractory gold ores and prevent longer-term distress due to grade erosion, we see three main areas for action that miners should consider:

— *Diligent mine planning and plant design are crucial to keep capital expenditures (capex) on budget and ensure that operating expenditures (opex) stay in the expected range during the production stage.* In order to decrease capex overspending, miners should pay extra attention to the plant design prior to construction. Identification of bottlenecks and overcapacities is crucial due to the many recirculation systems needed at the processing plant. A suboptimal plant design for a specific ore can quickly erode the benefits of higher-grade refractory ores through reduced recovery or the inability to approach nominal capacity. Miners should also be mindful of proper material selection during plant construction in order to decrease downtimes caused by, for example, extensive material wear in the highly corrosive environment associated with autoclaves.

— *Digital technology will help optimize throughput and yield at the plant and mitigate costs arising from grade erosion.* Custom-built artificial intelligence systems that use massive data generated by operations can significantly increase plant performance, as we have seen with clients that were able to achieve up to a 10 percent throughput increase and a 2 percent increase in recovery.[2]

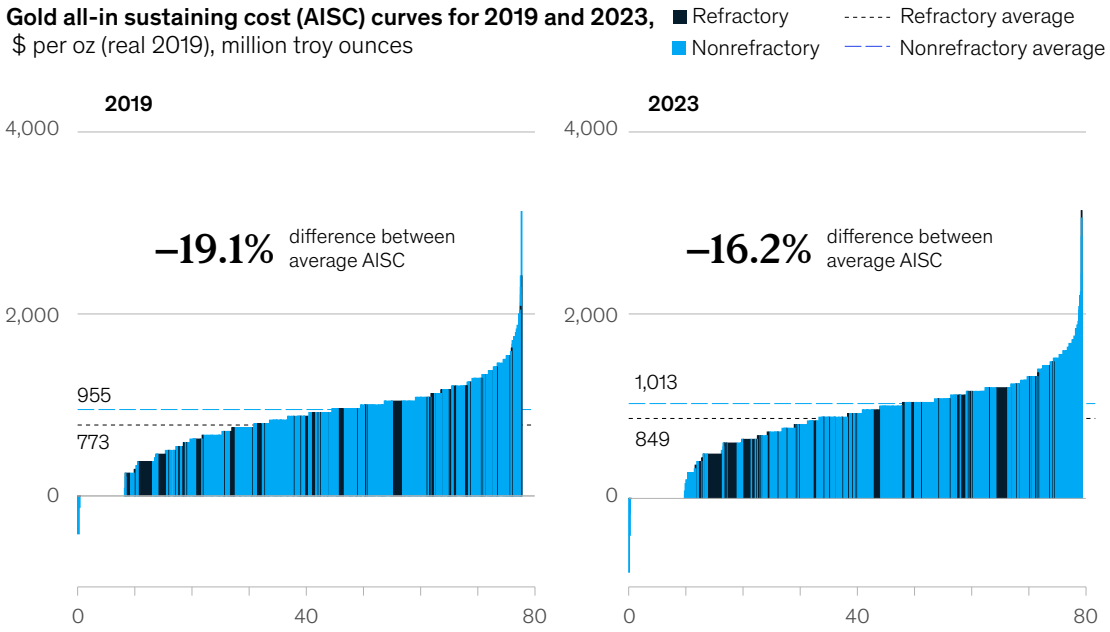
— *Inclusion of refractory processing plants in a miner's asset list may unlock the value of refractory-type reserves in other locations.*

[2]To learn more, see Red Conger, Harry Robinson, and Richard Sellschop, "Inside a mining company's AI transformation," February 2020, McKinsey.com.

Exhibit 5

Refractory ores yield costs per ounce that are often lower than average cost for conventional ores.

Gold all-in sustaining cost (AISC) curves for 2019 and 2023, $ per oz (real 2019), million troy ounces



[1]For example freight, mine overheads, etc.
Source: MineSpans

Miners could ship high-grade refractory ore or concentrate from several mines to centralized processing facilities designed to process difficult ores—an approach already employed by Russian miners Petropavlovsk and Polymetal International. Spare capacity at refractory-processing hubs can be used to process ore delivered from third-party miners that don't have their own refractory-processing circuit. Another benefit of having a refractory-ore processing plant is the transfer of know-how and experience between different operations.

Miners can tap into the opportunity for lower costs per ounce offered by higher-grade refractory gold reserves. However, when considering this potential, they need to factor in higher capital and operational costs. The winners will be those that pay close attention to optimizing mine design, maximizing plant performance via digital technology, and asset-portfolio planning.

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The authors wish to thank Jonathan Lindley for his contributions to this article.

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